

Tue, Apr 15

Newton's Raphson Method

approximate

maybe (nearest to zero)

Factor theorem

Non-linear Equation: $3x - \cos x = 1$

$$x^3 - 3x^2 + 5$$

$(x_0) \rightarrow$ interval

$$\frac{f(a) \cdot f(b)}{4} < 0$$

$$f(a) = 1$$

$$f(1) = 2.768$$

$$f(2) = -5.687 \quad [1, 2]$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad n = 0, 1, 2, 3, \dots$$

$$\textcircled{0} \quad 3x - \cos x = 1$$

find the exact real root correct up to 5 dp

$$f(x) = 3x - \cos x = 1$$

$$f(x) = 3x - \cos x - 1 = 0$$

$$f'(x) = 3 + \sin x \quad \therefore f'(x) \text{ should not be zero}$$

$$f(x) = 3x - \cos x - 1 = 0$$

$$f(0) = -2$$

$$f(1) = 1.45970$$

$[0, 1]$



$$x_0 = 0.6$$

$$n = 0$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.6 - \frac{3(0.6) - \cos(0.6) - 1}{3 + \sin(0.6)}$$

$$x_1 = 0.60711$$

$$x_2 = 0.60710$$

$$x_3 = 0.60710$$

Thursday Section A PDF

a) Find the real root of the following correct

upto 3 decimal places

$$f(x) = x^3 - 3x - 5 = 0$$

$$f'(x) = 3x^2 - 3$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(0) = -5$$

$$f(1) = -7$$

$$f(2) = -3$$

$$f(3) = 13$$

Intervals $[2, 3]$

$$\left. \begin{array}{l} f(2) = -3 \\ f(3) = 13 \end{array} \right\} f(a) \cdot f(b) < 0$$

$$f(2) = -3$$

$$f(2.1) = -2.039$$

$$f(2.2) = -0.952$$

$$f(2.3) = 0.2677$$

$[2.2, 2.3]$

$$f(2.27) = -0.112$$

$$f(2.28) = 0.0123$$

$[2.27, 2.28]$

$$f(2.279) = -0.0002$$

$$f(2.28) = 0.0123$$

closed $x = 2.279$

8) Find the root $f(x) = 3x - \cos x - 1 = 0$

correct upto 8 decimal points.

$$f(x) = 3x - \cos x - 1 = 0$$

$$f'(x) = 3 - (-\sin x) - 1 = 0$$

$$= 3 + \sin x - 1$$

$$f(0) = -2$$

$$f(1) = 1.4596$$

$$[0, 1]$$

$$f(0.6) = -0.029$$

$$f(0.7) = 0.3351$$

,

a) Find the real root of the following correct

upto 3 decimal places

$$f(x) = x^3 - 3x - 5 = 0$$

$$f'(x) = 3x^2 - 3$$

$$[2, 3]$$

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$$

$$x_0 = 2 \quad f(2) = \frac{(2)^3 - 3(2) - 5}{3(2)^2 - 3} = \frac{0 - 5}{-3} = 1.667$$

$$x_0 = 1.667$$

$$x_1 = 2.672$$

$$x_2 = 2.343$$

$$x_3 = 2.281$$

$$x_4 = 2.279$$

$$x_5 = 2.279$$

$$x_6 = 2.279$$

$$\underline{x = 2.279}$$

Q2) Find the real root of the following correct

upto 2 decimal places

$$f(x) = x^3 - 3x - 5 = 0$$

$$f'(x) = 3x^2 - 3$$

$$f(1) = -7$$

$$f(2) = -3$$

$$f(2.26) = -0.226$$

$$f(2) = -3$$

$$f(2.1) = -2.039$$

$$f(2.27) = -0.112$$

$$f(3) = 13$$

$$f(2.2) = -0.952$$

$$f(2.28) = 0.0123$$

$$[2, 3]$$

$$f(2.3) = 0.267$$

$$[2.2, 2.3]$$

$$[2.27, 2.28]$$

$$f(2.279) = -2 \times 10^{-4} \text{ or } 0.0002$$

$$f(2.280) = 0.0123$$



$$[2.279, 2.280]$$

$$x = 2.279$$

$$f(x) = 3x - \cos x = 1$$

$$f(x) = 3x - \cos x - 1 = 0$$

$$f'(x) = 3 - (-\sin x) + 0 = 0$$

$$f'(x) = 3 + \sin x$$

$$f(x) = 3x - \cos x - 1$$

$$f'(x) = 3 + \sin x$$

$$[0, 1]$$

$$x = 0$$

$$x_0 = 0 - \left(\frac{3(0) - \cos(0) - 1}{3 + \sin(0)} \right)$$

$$\text{Ans} = \left(\frac{3(\text{Ans}) - \cos(\text{Ans}) - 1}{3 + \sin(\text{Ans})} \right)$$

$$x_0 = 0.66667$$

$$x_1 = 0.60749$$

$$x_2 = 0.60710$$

$$x_3 = 0.60710$$

$$x = 0.60710$$

$$Q) \quad f(x) = x + \ln x - 2 \quad \text{sdp}$$

$$f(x) = x + \ln x - 2$$

$$f'(x) = 1 + \frac{1}{x}$$

$$[1, 2]$$

$$x = 1$$

$$x_0 = 1 - \left(\frac{1 + \ln(1) - 2}{1 + \frac{1}{1}} \right)$$

$$\text{Ans} = \left(\frac{\text{Ans} + \ln(\text{Ans}) - 2}{1 + \frac{1}{\text{Ans}}} \right)$$

$$x_0 = 1.5$$

$$x_1 = 1.55672$$

$$x_2 = 1.55715$$

$$x_3 = 1.55715$$

$$x = 1.55715$$

$$f(x) = e^{-x} - \sin x = 0$$

$$f'(x) = -(1)e^{-x} - \cos x = 0$$

$$f'(x) = -e^{-x} - \cos x = 0$$

$$f(x) = e^{-x} - \sin x$$

$$f'(x) = -e^{-x} - \cos x$$

$$[0, 1]$$

$$x = 0$$

$$x_0 = 0 - \left(\frac{e^{-0} - \sin(0)}{-e^{-0} - \cos(0)} \right)$$

$$x = \text{Ans} - \left(\frac{e^{-\text{Ans}} - \sin(\text{Ans})}{-e^{-\text{Ans}} - \cos(\text{Ans})} \right)$$

$$x_0 = 0.5$$

$$x_1 = 0.586$$

$$x_2 = 0.589$$

$$x_3 = 0.589$$

$$x = 0.589$$